1) a)

$e^{3x} = 1 + 3x + \frac{(3x)^2}{2!} + \frac{(3x)^3}{3!} + \cdots$

$\approx 1 + 3x + \frac{9}{2}x^2 + \frac{9}{2}x^3$ valid for all x

1) b)

$e^{-2x} = 1 - 2x + \frac{(-2x)^2}{2!} + \frac{(-2x)^3}{3!} + \cdots$

$\approx 1 - 2x + 2x^2 - \frac{4}{3}x^3$ valid for all x

1) c)

Validity: $-1 < 2x \leq 1 \Rightarrow -\frac{1}{2} < x \leq \frac{1}{2}$

$\ln(1 + 2x) = 2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} - \frac{(2x)^4}{4} + \cdots$

$\approx 2x - 2x^2 + \frac{8}{3}x^3 - 4x^4$ valid for $-\frac{1}{2} < x \leq \frac{1}{2}$

1) d)

Validity: $-1 < -x \leq 1 \Rightarrow -1 \leq x < 1$

$\ln(1 - x) = -x - \frac{(-x)^2}{2} + \frac{(-x)^3}{3} - \frac{(-x)^4}{4} + \cdots$

$\approx -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4$ valid for $-1 \leq x < 1$

1) e)

$\sin 2x = 2x - \frac{(2x)^3}{3!} - \frac{(2x)^5}{5!} - \frac{(2x)^7}{7!} + \cdots$

$\approx 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 - \frac{8}{315}x^7$ valid for all x

1) f)

$\sin(-4x) = -4x - \frac{(-4x)^3}{3!} - \frac{(-4x)^5}{5!} - \frac{(-4x)^7}{7!} + \cdots$

$\approx -4x - \frac{32}{3}x^3 - \frac{128}{15}x^5 + \frac{1024}{315}x^7$ valid for all x

1) g)

$\cos\left(\frac{1}{2}x\right) = 1 - \frac{\left(\frac{1}{2}x\right)^2}{2!} + \frac{\left(\frac{1}{2}x\right)^4}{4!} - \frac{\left(\frac{1}{2}x\right)^6}{6!} + \cdots$

$\approx 1 - \frac{1}{8}x^2 + \frac{1}{384}x^4 - \frac{1}{46080}x^6$ valid for all x

1) h)

$\cos(-x) = 1 - \frac{(-x)^2}{2!} + \frac{(-x)^4}{4!} - \frac{(-x)^6}{6!} + \cdots$

$\approx 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6$ valid for all x

2) a)

Second term: $-\frac{\left(\frac{5}{2}x\right)^2}{2!}$

Coefficient: $-\frac{25}{8}$

2) b)

Second term: $-4x$

Coefficient: -4

2) c)

Second term: $-\frac{\left(\frac{1}{2}x\right)^2}{2}$

Coefficient: $-\frac{1}{8}$

3) a)

$\ln(3+x) = \ln\left(3\left(1+\frac{1}{3}x\right)\right) = \ln 3 + \ln\left(1+\frac{1}{3}x\right)$

$\ln(1-x) \approx \ln 3 + \frac{1}{3}x - \frac{\left(\frac{1}{3}x\right)^2}{2} + \cdots$

$\approx \ln 3 + \frac{1}{3}x - \frac{1}{18}x^2$

3) b)

Valid for $-1 < \frac{1}{3}x \leq 1$

So $-3 < x \leq 3$

4) a)

$$\ln\left(1+\frac{1}{4}x\right) = \left(\frac{1}{4}x\right) - \frac{\left(\frac{1}{4}x\right)^2}{2} + \frac{\left(\frac{1}{4}x\right)^3}{3} + \cdots$$
$$\approx \frac{1}{4}x - \frac{1}{32}x + \frac{1}{192}x^3$$

4) b)

$-1 < \frac{1}{4}x \leq 1$ so $-4 < x \leq 4$

4) c)

$$\ln\sqrt{1+\frac{1}{4}x} = \ln\left(1+\frac{1}{4}x\right)^{\frac{1}{2}} = \frac{1}{2}\ln\left(1+\frac{1}{3}x\right) \approx \frac{1}{8}x - \frac{1}{64}x + \frac{1}{384}x^3$$

5)

$\frac{p^5}{5!} = \frac{4}{15} \Rightarrow p^5 = 32$

$\therefore p = 2$

6)

$\frac{p^2}{2!} = 72$

$p^2 = 144$

So $p = \pm 12$

7) a)

$\ln(4(1+x)) = \ln 4 + \ln(1+x)$

$\approx \ln 4 + x - \frac{x^2}{2} + \frac{x^3}{3}$

7) b)

$\frac{3e^x+1}{e^x} = 3 + e^{-x}$

$\approx 3 + 1 + (-x) + \frac{(-x)^2}{2!} + \frac{(-x)^3}{3!}$

$\approx 4 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3$

7) c)

$$\ln(\mathrm{e}-2\mathrm{e}x)=\ln\big(\mathrm{e}(1-2x)\big)$$
$$\approx \ln(\mathrm{e})+\ln(1-2x)$$
$$\approx \ln(\mathrm{e})+(-2x)-\frac{(-2x)^2}{2}+\frac{(-2x)^3}{3}$$
$$\approx 1-2x-2x^2-\frac{8}{3}x^3$$

7) d)

$$\ln(1+x^2)=x^2-\frac{(x^2)^2}{2!}+\frac{(x^2)^3}{3!}\approx x^2-\frac{x^4}{2}+\frac{x^2}{6}$$
$$\frac{1}{x}\ln(1+x^2)\approx x-\frac{x^3}{2}$$

7) e)

$$\mathrm{e}^{x^2}=1+(x^2)+\frac{(x^2)^2}{2!}+\cdots\approx 1+x^2+\frac{1}{2}x^4$$
$$\frac{1}{x}\mathrm{e}^{x^2}\approx\frac{1}{x}+x+\frac{1}{2}x^3$$

7) f)

$$\ln(2-2x)=\ln\big(2(1-x)\big)=\ln 2+\ln(1-x)$$
$$\ln(1-x)=-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}-\cdots$$
$$\ln(2-2x)=\ln 2-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}-\cdots$$
$$\frac{1}{x}\ln(2-2x)=\frac{\ln 2}{x}-1-\frac{x}{2}-\frac{x^2}{3}-\frac{x^3}{4}-\cdots$$
$$\approx\frac{\ln 2}{x}-1-\frac{x}{2}-\frac{x^2}{3}-\frac{x^3}{4}$$

8) a)

$$\cos(2x)\approx 1-\frac{(2x)^2}{2!}+\frac{(2x)^4}{4!}-\frac{(2x)^6}{6!}+\frac{(2x)^8}{8!}$$
$$\approx 1-2x^2+\frac{2}{3}x^4-\frac{4}{45}x^6+\frac{2}{315}x^8$$

8) b)

$$\cos(-2x)\approx 1-2x^2+\frac{2}{3}x^4-\frac{4}{45}x^6+\frac{2}{315}x^8$$

9) a)

$\sin(2x) \approx 2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \frac{(2x)^7}{7!}$

$\approx 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 - \frac{8}{315}x^7$

9) b)

$\sin(-2x) \approx -2x + \frac{4}{3}x^3 - \frac{4}{15}x^5 + \frac{8}{315}x^7$

10) a)

$(4+4x)^{\frac{1}{2}} = (4(1+x))^{\frac{1}{2}} = 4^{\frac{1}{2}}(1+x)^{\frac{1}{2}}$

$= 2\left[1 + \frac{1}{2}x + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)x^2}{2} + \cdots\right]$

$= 2 + x - \frac{1}{4}x^2 + \cdots$ for $|x| < 1$

10) b)

$(4+x)^{\frac{1}{2}} = \left(4\left(1+\frac{1}{4}x\right)\right)^{\frac{1}{2}} = 4^{\frac{1}{2}}\left(1+\frac{1}{4}x\right)^{\frac{1}{2}}$

$= 2\left[1 + \frac{1}{2}\left(\frac{1}{4}x\right) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(\frac{1}{4}x\right)^2}{2} + \cdots\right]$

$= 2 + \frac{1}{4}x - \frac{1}{64}x^2 + \cdots$ for $\left|\frac{1}{4}x\right| < 1$

So $(4+x)^{\frac{1}{2}} \approx 2 + \frac{1}{4}x - \frac{1}{64}x^2$ for $|x| < 4$

10) c)

$(4+5x)^{-\frac{1}{2}} = \left(4\left(1+\frac{5}{4}x\right)\right)^{-\frac{1}{2}} = 4^{-\frac{1}{2}}\left(1+\frac{5}{4}x\right)^{-\frac{1}{2}}$

$= \frac{1}{2}\left[1 + \left(-\frac{1}{2}\right)\left(\frac{5}{4}x\right) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(\frac{5}{4}x\right)^2}{2} + \cdots\right]$

$= \frac{1}{2} - \frac{5}{16}x + \frac{75}{256}x^2 + \cdots$ for $\left|\frac{5}{4}x\right| < 1$

So $(4+5x)^{-\frac{1}{2}} \approx \frac{1}{2} - \frac{5}{16}x + \frac{75}{256}x^2$ for $|x| < \frac{4}{5}$

11) a)

$\ln(1+3x^2-x) = (3x^2-x) - \frac{(3x^2-x)^2}{2} + \frac{(3x^2-x)^3}{3} \ldots$

$= 3x^2 - x - \frac{9x^4-6x^3+x^2}{2} + \frac{(3x^2-x)(9x^4-6x^3+x^2)}{3} \ldots$

$= 3x^2 - x + 3x^3 - \frac{1}{2}x^2 - \frac{1}{3}x^3 \ldots$

$\approx -x + \frac{5}{2}x^2 + \frac{8}{3}x^3$

$-1 < 3x^2 - x \leq 1$

$3x^2 - x \leq 1 \Rightarrow 3x^2 - x - 1 \leq 0 \Rightarrow \frac{1}{6} - \frac{\sqrt{13}}{6} \leq x \leq \frac{1}{6} + \frac{\sqrt{13}}{6}$

$-1 < 3x^2 - x \Rightarrow 3x^2 - x + 1 > 0 \Rightarrow 3\left(x - \frac{1}{6}\right)^2 + \frac{11}{12} > 0$

which is always true

So $\ln(1+3x^2-x) \approx -x + \frac{5}{2}x^2 + \frac{8}{3}x^3$ where $\frac{1}{6} - \frac{\sqrt{13}}{6} \leq x \leq \frac{1}{6} + \frac{\sqrt{13}}{6}$

11) b)

$\ln(1+2x^2+3x) = (2x^2+3x) - \frac{(2x^2+3x)^2}{2} + \frac{(2x^2+3x)^3}{3} + \cdots$

$= 2x^2 + 3x - \frac{4x^4+12x^3+9x^2}{2} + \frac{(2x^2+3x)(4x^4+12x^3+9x^2)}{3} + \cdots$

$= 2x^2 + 3x - 6x^3 - \frac{9}{2}x^2 + 9x^3 \ldots$

$\approx 3x - \frac{5}{2}x^2 + 3x^3$

$-1 < 2x^2 + 3x \leq 1$

$2x^2 + 3x \leq 1 \Rightarrow 2x^2 + 3x - 1 \leq 0 \Rightarrow -\frac{3}{4} - \frac{\sqrt{17}}{4} \leq x \leq -\frac{3}{4} + \frac{\sqrt{17}}{4}$

$-1 < 2x^2 + 3x \Rightarrow 2x^2 + 3x + 1 > 0 \Rightarrow x > -\frac{1}{2}, x < -1$

So $\ln(1+2x^2+3x) \approx 3x - \frac{5}{2}x^2 + 3x^3$

where $-\frac{3}{4} - \frac{\sqrt{17}}{4} \leq x < -1$ or $-\frac{1}{2} < x \leq -\frac{3}{4} + \frac{\sqrt{17}}{4}$

12) a)

$\ln\left(1 + \frac{1}{2}x^2 + 4x\right) = \left(\frac{1}{2}x^2 + 4x\right) - \frac{\left(\frac{1}{2}x^2+4x\right)^2}{2} + \frac{\left(\frac{1}{2}x^2+4x\right)^3}{3} + \cdots$

$\frac{1}{2}x^2 + 4x - \frac{\frac{1}{4}x^4+4x^3+16x^2}{2} + \frac{\left(\frac{1}{2}x^2+4x\right)\left(\frac{1}{4}x^4+4x^3+16x^2\right)}{3} + \cdots$

$\frac{1}{2}x^2 + 4x - 2x^3 - 8x^2 + \frac{64}{3}x^3 + \cdots$

$\approx 4x - \frac{15}{2}x^2 + \frac{58}{3}x^3$

12) b)

$$\ln(x^2+8x+2)=\ln\left(2\left(\frac{1}{2}x^2+4x+1\right)\right)$$
$$=\ln 2+\ln\left(1+\frac{1}{2}x^2+4x\right)$$
$$\approx \ln 2+4x-\frac{15}{2}x^2+\frac{58}{3}x^3$$

13) a)

$$e^x+e^{-x}=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots+1-x+\frac{x^2}{2!}-\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots$$
$$=1+x+\frac{x^2}{2}+\frac{x^3}{6}+\frac{x^4}{24}+\cdots+1-x+\frac{x^2}{2}-\frac{x^3}{6}+\frac{x^4}{24}+\cdots$$
$$\approx 2+x^2+\frac{1}{12}x^4$$

13) b)

$$\sin x+\cos x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots+1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots$$
$$\approx 1+x-\frac{x^2}{2}$$

13) c)

$$\cos x-\sin x=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots-x+\frac{x^3}{3!}-\frac{x^5}{5!}-\cdots$$
$$\approx 1-x-\frac{x^2}{2}$$

13) d)

$$e^x-\sin x=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots-x+\frac{x^3}{3!}-\frac{x^5}{5!}+\cdots$$
$$\approx 1+\frac{x^2}{2}+\frac{x^3}{3}$$

13) e)

$$e^x-\cos x=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots-1+\frac{x^2}{2!}-\frac{x^4}{4!}-\cdots$$
$$\approx x+x^2+\frac{x^3}{6}$$

13) f)

$$2\sin 2x + 2\cos 2x = 2\left(2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \cdots\right) + 2\left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \cdots\right)$$
$$= 2\left(2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 - \cdots\right) + 2\left(1 - 2x^2 + \frac{2}{3}x^4 - \cdots\right)$$
$$= 4x - \frac{8}{3}x^3 + \frac{8}{15}x^5 - \cdots + 2 - 4x^2 + \frac{4}{3}x^4 - \cdots$$
$$\approx 2 + 4x - 4x^2$$

13) g)

$$\mathrm{e}^{ax} + \mathrm{e}^{-ax} = 1 + ax + \frac{(ax)^2}{2!} + \frac{(ax)^3}{3!} + \frac{(ax)^4}{4!} + \cdots + 1 - ax + \frac{(ax)^2}{2!} - \frac{(ax)^3}{3!} + \frac{(ax)^4}{4!} + \cdots$$
$$= 1 + ax + \frac{a^2x^2}{2} + \frac{a^3x^3}{6} + \frac{a^4x^4}{24} + \cdots + 1 - ax + \frac{a^2x^2}{2} - \frac{a^3x^3}{6} + \frac{a^4x^4}{24} + \cdots$$
$$\approx 2 + a^2x^2 + \frac{a^4}{12}x^4$$

13) h)

$$\sin ax + x\cos ax = ax - \frac{(ax)^3}{3!} + \frac{(ax)^5}{5!} - \cdots + x\left(1 - \frac{(ax)^2}{2!} + \frac{(ax)^4}{4!} - \cdots\right)$$
$$= ax - \frac{a^3}{6}x^3 + \frac{(ax)^5}{5!} \ldots + x\left(1 - \frac{a^2}{2}x^2 + \frac{a^4}{24}x^4 - \cdots\right)$$
$$= ax - \frac{a^3}{6}x^3 + \frac{a^5}{120}x^5 \ldots + x - \frac{a^2}{2}x^3 + \frac{a^4}{24}x^5 - \cdots$$
$$\approx (1+a)x - \frac{a^2}{6}(a+3)x^3 + \frac{1}{120}a^4(a+5)x^5$$

14) a)

$$|x| < \frac{1}{b}$$

14) b)

$$\mathrm{f}(x) = 1 + ax + \frac{(ax)^2}{2!} + \frac{(ax)^3}{3!} + \cdots + 1 + \left(\frac{1}{2}\right)(bx) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)(bx)^2}{2} + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(bx)^3}{6} + \cdots$$
$$\approx 1 + ax + \frac{a^2}{2}x^2 + \frac{a^3}{6}x^3 + 1 + \frac{b}{2}x - \frac{b^2}{8}x^2 + \frac{b^3}{16}x^3$$
$$= 2 + \left(a + \frac{b}{2}\right)x + \left(\frac{a^2}{2} - \frac{b^2}{8}\right)x^2 + \left(\frac{a^3}{6} + \frac{b^3}{16}\right)x^3$$

14) c)

$a + \frac{b}{2} = -1$ and $\frac{a^2}{2} - \frac{b^2}{8} = \frac{5}{2}$

So $b = -2a - 2$ which gives

$\frac{a^2}{2} - \frac{(-2a-2)^2}{8} = \frac{5}{2} \Rightarrow 4a^2 - (-2a-2)^2 = 20$

$4a^2 - (4a^2 + 8a + 4) = 20$

$-8a = 24$

so $a = -3$ and $b = 4$

14) d)

$\frac{a^3}{6} + \frac{b^3}{48} = \frac{(-3)^3}{6} + \frac{4^3}{16} = -\frac{1}{2}$

15)

$e^{2x} = 1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \frac{(2x)^4}{4!} + \frac{(2x)^5}{5!} + \cdots$

$= 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \frac{4}{15}x^5 + \cdots$

$e^{-2x} = 1 - 2x + 2x^2 - \frac{4}{3}x^3 + \frac{2}{3}x^4 - \frac{4}{15}x^5 + \cdots$

$\sinh(2x) = \frac{1}{2}(e^{2x} - e^{-2x})$

$= \frac{1}{2}\left(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \frac{4}{15}x^5 + \cdots - \left(1 - 2x + 2x^2 - \frac{4}{3}x^3 + \frac{2}{3}x^4 - \frac{4}{15}x^5 + \cdots\right)\right)$

$= \frac{1}{2}\left(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \frac{4}{15}x^5 + \cdots - 1 + 2x - 2x^2 + \frac{4}{3}x^3 - \frac{2}{3}x^4 + \frac{4}{15}x^5 + \cdots\right)$

$\approx \frac{1}{2}\left(4x + \frac{8}{3}x^3 + \frac{8}{15}x^5\right)$

$\approx 2x + \frac{4}{3}x^3 + \frac{4}{15}x^5$

16)

$e^{2x} = 1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \frac{(2x)^4}{4!} + \cdots$

$= 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \cdots$

$e^{-2x} = 1 - 2x + 2x^2 - \frac{4}{3}x^3 + \frac{2}{3}x^4 + \cdots$

$\cosh(2x) = \frac{1}{2}(e^{2x} - e^{-2x})$

$= \frac{1}{2}\left(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \cdots + 1 - 2x + 2x^2 - \frac{4}{3}x^3 + \frac{2}{3}x^4 - \cdots\right)$

$\approx \frac{1}{2}\left(2 + 4x^2 + \frac{4}{3}x^4\right)$

$\approx 1 + 2x^2 + \frac{2}{3}x^4$

17) a)

$e^{ix} = 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \cdots$

$= 1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \cdots$

$e^{-ix} = 1 - ix + \frac{(-ix)^2}{2!} + \frac{(-ix)^3}{3!} + \frac{(-ix)^4}{4!} + \cdots$

$= 1 - ix - \frac{1}{2}x^2 + \frac{i}{6}x^3 + \frac{1}{24}x^4 + \cdots$

$\cosh(ix) = \frac{1}{2}\left(e^{ix} + e^{-ix}\right)$

$= \frac{1}{2}\left(1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \cdots + 1 - ix - \frac{1}{2}x^2 + \frac{i}{6}x^3 + \frac{1}{24}x^4 + \cdots\right)$

$= \frac{1}{2}\left(2 - x^2 + \frac{1}{12}x^4 + \cdots\right)$

$\approx 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

17) b)

$1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 = 1 - \frac{x^2}{2!} + \frac{x^4}{6!} = \cos x$

$\therefore \cos x$

18) a)

$e^{ix} = 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \cdots$

$= 1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \frac{i}{120}x^5 + \cdots$

$e^{-ix} = 1 - ix + \frac{(-ix)^2}{2!} + \frac{(-ix)^3}{3!} + \frac{(-ix)^4}{4!} + \frac{(-ix)^5}{5!} + \cdots$

$= 1 - ix - \frac{1}{2}x^2 + \frac{i}{6}x^3 + \frac{1}{24}x^4 - \frac{i}{120}x^5 + \cdots$

$\sinh(ix) = \frac{1}{2}\left(e^{ix} + e^{-ix}\right)$

$= \frac{1}{2}\left(1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \frac{i}{120}x^5 + \cdots - \left(1 - ix - \frac{1}{2}x^2 + \frac{i}{6}x^3 + \frac{1}{24}x^4 - \frac{i}{120}x^5 + \cdots\right)\right)$

$= \frac{1}{2}\left(1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \frac{i}{120}x^5 + \cdots - 1 + ix + \frac{1}{2}x^2 - \frac{i}{6}x^3 - \frac{1}{24}x^4 + \frac{i}{120}x^5 + \cdots\right)$

$\approx \frac{1}{2}\left(2ix - \frac{i}{3}x^3 + \frac{i}{60}x^5\right)$

$\approx ix - \frac{i}{6}x^3 + \frac{i}{120}x^5$

18) b)

$ix - \frac{i}{6}x^3 + \frac{i}{120}x^5 = i\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5\right)$

$\therefore \sin x$

19) a)

$(1 + 2x)^{\frac{1}{3}} = 1 + \left(\frac{1}{3}\right)(2x) + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)(2x)^2}{2} + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)(2x)^3}{6} + \cdots$

$= 1 + \frac{2}{3}x - \frac{4}{9}x^2 + \frac{40}{81}x^3 + \cdots$

$\ln(1 + 5x) = (5x) - \frac{(5x)^2}{2} + \frac{(5x)^3}{3} + \cdots = 5x - \frac{25}{2}x^2 + \frac{125}{3}x^3 + \cdots$

$k(1 + 2x)^{\frac{1}{3}} - \left(1 + \frac{2}{3}x - \frac{4}{9}x^2 + \frac{40}{81}x^3 + \cdots\right)$

$= k\left(1 + \frac{2}{3}x - \frac{4}{9}x^2 + \frac{40}{81}x^3 + \cdots\right) - \left(5x - \frac{25}{2}x^2 + \frac{125}{3}x^3 + \cdots\right)$

$\frac{2}{3}k - 5 = -1$ so $k = 6$

19) b)

$6\left(1+\frac{2}{3}x-\frac{4}{9}x^2+\frac{40}{81}x^3+\cdots\right)-\left(5x-\frac{25}{2}x^2+\frac{125}{3}x^3+\cdots\right)$

$=6+4x-\frac{8}{3}x^2+\frac{80}{27}x^3+\cdots-5x+\frac{25}{2}x^2-\frac{125}{3}x^3-\cdots$

$\approx 6-x+\frac{59}{6}x^2-\frac{1045}{27}x^3$

19) c)

$(1+2x)^{\frac{1}{3}}$ is valid for $-\frac{1}{2}<x\leq\frac{1}{2}$

$\ln(1+5x)$ is valid for $-\frac{1}{5}<x\leq\frac{1}{5}$

So $f(x)$ is valid for $-\frac{1}{5}<x\leq\frac{1}{5}$

20) a)

$(1-2x-8x^2)=(1+2x)(1-4x)$

20) b)

$\ln(1-2x-8x^2)$

$=\ln\big((1+2x)(1-4x)\big)$

$=\ln(1+2x)+\ln(1-4x)$

$=\left(2x-2x^2+\frac{8}{3}x^3-\right)\ldots+\left(-4x-8x^2-\frac{64}{3}x^3-\cdots\right)$

$\approx -2x-10x^2-\frac{56}{3}x^3$

20) c)

$\ln(1+2x)$ is valid for $-1<2x\leq 1\therefore -\frac{1}{2}<x\leq\frac{1}{2}$

$\ln(1-4x)$ is valid for $-1<-4x\leq 1$

$-1<-4x$ gives $x<\frac{1}{4}$ and $-4x\leq 1$ gives $x\geq-\frac{1}{4}$ so $-\frac{1}{4}\leq x<\frac{1}{4}$

So $\ln(1-2x-8x^2)$ is valid for $-\frac{1}{4}\leq x<\frac{1}{4}$

21) a)

$$e^x \sin x = \left(1 + x + \frac{x^2}{2!} + \cdots\right)\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)$$
$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \cdots + \frac{x^3}{2} - \frac{x^5}{2\times 3!} + \frac{x^7}{2\times 5!} + \cdots$$
$$\approx x + x^2 - \frac{x^3}{3!} + \frac{x^3}{2} + \cdots$$
$$\approx x + x^2 + \frac{1}{3}x^3$$

21) b)

$$e^x \cos x = \left(1 + x + \frac{x^2}{2!} + \cdots\right)\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right)$$
$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + x - \frac{x^3}{2!} + \frac{x^5}{4!} - \cdots + \frac{x^2}{2!} - \frac{x^4}{2\times 2!} + \frac{x^6}{2\times 4!} - \cdots$$
$$\approx 1 + x - \frac{1}{2}x^3$$

21) c)

$$\sin x \ln(1 + x) = \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots\right)$$
$$= x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \cdots - \frac{x^4}{6} + \frac{x^5}{6\times 2} - \frac{x^6}{6\times 3} + \cdots$$
$$\approx x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \frac{x^4}{6} \ldots$$
$$\approx x^2 - \frac{1}{2}x^3 + \frac{1}{6}x^4$$

21) d)

$$e^{-x} \sin x = \left(1 - x + \frac{x^2}{2!} + \cdots\right)\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)$$
$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots - x^2 + \frac{x^4}{3!} - \frac{x^6}{5!} - \cdots + \frac{x^3}{2} - \frac{x^5}{2\times 3!} + \frac{x^7}{2\times 5!} + \cdots$$
$$\approx x - \frac{1}{6}x^3 - x^2 + \frac{1}{2}x^3$$
$$\approx x - x^2 + \frac{1}{3}x^3$$

21) e)

$$e^x \cos 2x = \left(1 + x + \frac{x^2}{2!} + \cdots\right)\left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \cdots\right)$$
$$= \left(1 + x + \frac{x^2}{2!} + \cdots\right)\left(1 - 2x^2 + \frac{2}{3}x^4 - \cdots\right)$$
$$= 1 - 2x^2 + \frac{2}{3}x^4 - \cdots + x - 2x^3 + \frac{2}{3}x^5 - \cdots + \frac{1}{2}x^2 - x^4 + \frac{2}{3}x^6 - \cdots$$
$$\approx 1 + x - \frac{3}{2}x^2$$

22) a)

$$(3 + 5x)^{-1} = \left(3\left(1 + \frac{5}{3}x\right)\right)^{-1} \approx \frac{1}{3}\left[1 + (-1)\left(\frac{5}{3}x\right) + \frac{(-1)(-2)\left(\frac{5}{3}x\right)^2}{2} + \frac{(-1)(-2)(-3)\left(\frac{5}{3}x\right)^3}{6} + \cdots\right]$$
$$= \frac{1}{3}\left[1 - \frac{5}{3}x + \frac{25}{9}x^2 - \frac{125}{27}x^3 + \cdots\right]$$
$$\approx \frac{1}{3} - \frac{5}{9}x + \frac{25}{27}x^2 - \frac{125}{81}x^3$$

22) b)

Valid for $\left|\frac{5}{3}x\right| < 1$ so $|x| < \frac{3}{5}$

22) c)

$$\sin 2x = 2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \cdots$$
$$= 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 - \cdots$$

So that

$$(\sin 2x)(3 + 5x)^{-1} = \left(2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 - \cdots\right)\left(\frac{1}{3} - \frac{5}{9}x + \frac{25}{27}x^2 - \frac{125}{81}x^3\right)$$
$$= \frac{2}{3}x - \frac{10}{9}x^2 + \frac{50}{27}x^3 - \frac{250}{81}x^4 + \cdots - \frac{4}{9}x^3 + \frac{20}{27}x^4 + \cdots$$
$$\approx \frac{2}{3}x - \frac{10}{9}x^2 + \frac{38}{27}x^3 - \frac{190}{81}x^4$$

22) d)

$$|x| < \frac{3}{5}$$

23) a)

$\cos\theta \sin\theta = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right)\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)$

$= \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots - \frac{\theta^3}{2} + \frac{\theta^5}{2\times3!} - \frac{\theta^7}{2\times5!} - \cdots + \frac{\theta^5}{24} - \frac{\theta^7}{24\times3!} + \frac{\theta^9}{24\times5!} - \cdots$

$\approx \theta - \frac{\theta^3}{6} + \frac{\theta^5}{120} - \frac{\theta^3}{2} + \frac{\theta^5}{12} + \frac{\theta^5}{24}$

$\approx \theta - \frac{2}{3}\theta^3 + \frac{2}{15}\theta^5$

$\sin q\theta = q\theta - \frac{(q\theta)^3}{3!} - \frac{(q\theta)^5}{5!} + \cdots$

$\approx q\theta - \frac{q^3}{6}\theta^3 + \frac{q^5}{120}\theta^5$

So $\cos\theta \sin\theta \approx \theta - \frac{2}{3}\theta^3 + \frac{2}{15}\theta^5$ and $\sin q\theta \approx q\theta - \frac{q^3}{6}\theta^3 + \frac{q^5}{120}\theta^5$

23) b)

$\theta - \frac{2}{3}\theta^3 + \frac{2}{15}\theta^5 = pq\theta - \frac{pq^3}{6}\theta^3 + \frac{pq^5}{120}\theta^5$

$pq = 1$

$-\frac{2}{3} = -\frac{pq^3}{6} \Rightarrow 4 = pq^3 \Rightarrow 4 = q^2 \therefore q = 2$

$p = \frac{1}{2}$ and $q = 2$

24)

$e^{px}\sin x = \left(1 + px + \frac{p^2x^2}{2!} + \cdots\right)\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)$

$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + px^2 - \frac{px^4}{3!} + \frac{px^6}{5!} - \cdots + \frac{p^2}{2}x^3 - \cdots$

So the third term is

$-\frac{1}{6}x^3 + \frac{p^2}{2}x^3$

$\frac{p^2}{2} - \frac{1}{6} = \frac{13}{3} \Rightarrow p^2 = 9$

$\therefore p = -3$ or $p = 3$

25)

$$e^x \sin px = \left(1 + x + \frac{x^2}{2!} + \cdots\right)\left(px - \frac{p^3x^3}{3!} + \frac{p^5x^5}{5!} - \cdots\right)$$

$$= px - \frac{p^3x^3}{3!} + \frac{p^5x^5}{5!} - \cdots + px^2 - \frac{p^3x^4}{3!} + \frac{p^5x^6}{5!} - \cdots + \frac{p}{2!}x^3 - \cdots$$

So the third term is

$$-\frac{p^3}{6}x^3 + \frac{p}{2}x^3$$

$$-\frac{p^3}{6} + \frac{p}{2} = -\frac{1}{3} \Rightarrow p^3 - 3p = 2 \Rightarrow p^3 - 3p - 2 = 0 \Rightarrow (p-2)(p+1)^2 = 0$$

$\therefore p = 2$ or $p = -1$

26)

$$\frac{\cos px}{e^x} = e^{-x}\cos px = \left(1 - x + \frac{x^2}{2!} + \cdots\right)\left(1 - \frac{(px)^2}{2!} + \frac{(px)^4}{4!} - \cdots\right)$$

$$= 1 - \frac{p^2x^2}{2!} + \frac{p^4x^4}{4!} - \cdots - x + \frac{p^2x^3}{2!} + \cdots + \frac{x^2}{2!} - \cdots$$

So the third term is

$$-\frac{p^2x^2}{2} + \frac{x^2}{2}$$

$$-\frac{p^2}{2} + \frac{1}{2} = -12 \Rightarrow p^2 - 1 = 24$$

$\therefore p = -5$ or $p = 5$

27)

$$\frac{\sin ax}{be^x} = \frac{1}{b}\sin ax\, e^{-x}$$

$$= \frac{1}{b}\left(ax - \frac{a^3x^3}{3!} + \frac{a^5x^5}{5!} - \cdots\right)\left(1 - x + \frac{x^2}{2!} + \cdots\right)$$

$$= \frac{1}{b}\left(ax - ax^2 + \frac{ax^3}{2} + \cdots - \frac{a^3x^3}{6} + \frac{a^3x^4}{6} - \frac{a^3x^5}{12} + \cdots\right)$$

$\therefore \frac{a}{b} = \frac{1}{2}$ and $\frac{1}{b}\left(\frac{a}{2} - \frac{a^3}{6}\right) = -\frac{1}{12}$

$$\frac{a}{2b} - \frac{a^3}{6b} = -\frac{1}{12} \Rightarrow \frac{1}{4} - \frac{a^2}{12} = -\frac{1}{12} \Rightarrow a^2 = 4$$

The second term is $-\frac{a}{b}$

$\therefore a = 2$ and $b = 4$ and $k = \frac{1}{2}$

28) a)

$e^{0.1} \approx 1 + 0.1 + \frac{(0.1)^2}{2!} + \frac{(0.1)^3}{3!} + \cdots$

$\approx 1 + 0.1 + 0.005 + 0.0001666666 + \cdots$

$= 1.1052$ to 4 decimal places

28) b)

$\ln 0.9 = \ln(1 - 0.1)$

$\approx -0.1 - \frac{(-0.1)^2}{2} + \frac{(-0.1)^3}{3} - \frac{(-0.1)^4}{4} + \cdots$

$\approx -0.1 - 0.005 - 0.00033333 \ldots - 0.000025 + \cdots$

$= -0.1054$ to 4 decimal places

28) c)

$\sin\left(\frac{\pi}{18}\right) \approx \frac{\pi}{18} - \frac{\left(\frac{\pi}{18}\right)^3}{3!} + \cdots$

$\approx 0.17453292 \ldots - 0.00088609 \ldots + \cdots$

$= 0.1736$ to 4 decimal places

28) d)

$14° = \frac{14\pi}{180}$ radians $= \frac{7\pi}{90}$ radians

$\cos\left(\frac{7\pi}{90}\right) \approx 1 - \frac{\left(\frac{7\pi}{90}\right)^2}{2!} + \frac{\left(\frac{7\pi}{90}\right)^4}{4!} + \cdots$

$\approx 1 - 0.02985250 \ldots + 0.00014852 + \cdots$

$= 0.9703$ to 4 decimal places

29) a)

$0.993252 \ldots$

29) b)

The expansion is only valid for $-1 < x \leq 1$

29) c)

$\ln(2.7) = \ln(3 \times 0.9) = \ln 3 + \ln 0.9 = \ln 3 + \ln(1 - 0.1)$

$\approx \ln 3 - 0.1 - \frac{(-0.1)^2}{2} + \frac{(-0.1)^3}{3} + \cdots$

$\approx 1.09861 - 0.1 - 0.005 - 0.00033333 + \cdots$

$=$ 0.9933 to 4 decimal places

30) a)

$\frac{1}{\sqrt{e^x}} = e^{-\frac{1}{2}x} = 1 + \left(-\frac{1}{2}x\right) + \frac{\left(-\frac{1}{2}x\right)^2}{2!} + \frac{\left(-\frac{1}{2}x\right)^3}{3!} + \frac{\left(-\frac{1}{2}x\right)^4}{4!} + \frac{\left(-\frac{1}{5}x\right)^5}{5!} \ldots$

$\approx 1 - \frac{1}{2}x + \frac{1}{8}x^2 - \frac{1}{48}x^3 + \frac{1}{384}x^4 - \frac{1}{3840}x^5 \ldots$

30) b)

$\frac{1}{\sqrt{e^1}} \approx 1 - \frac{1}{2} + \frac{1}{8} - \frac{1}{48} + \frac{1}{384} - \frac{1}{3840} + \cdots$

$= 0.6065$

31) a)

$e^x \approx 1 + x + \frac{x^2}{2!} \approx 1 + x + \frac{1}{2}x^2$

$\int_0^{0.5} e^x \, dx \approx \int_0^{0.5} \left(1 + x + \frac{1}{2}x^2\right) dx$

$= \left[x + \frac{1}{2}x^2 + \frac{1}{6}x^3\right]_0^{0.5}$

$= \left(0.5 + \frac{1}{2} \times 0.5^2 + \frac{1}{6} \times 0.5^3\right) - (0)$

$=$ 0.65 (2 DP)

31) b)

$\ln(1 + x) \approx x - \frac{x^2}{2} + \frac{x^3}{3}$

$\int_0^{0.5} \ln(1 + x) \, dx \approx \int_0^{0.5} \left(x - \frac{x^2}{2} + \frac{x^3}{3}\right) dx$

$= \left[\frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4\right]_0^{0.5}$

$= \left(\frac{1}{2} \times 0.5^2 - \frac{1}{6} \times 0.5^3 + \frac{1}{12} \times 0.5^4\right) - (0)$

$=$ 0.11 (2 DP)

31) c)

$\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} = x - \frac{1}{6}x^3 + \frac{1}{120}x^5$

$\int_0^{1.5} \sin x \, dx \approx \int_0^{1.5} \left(x - \frac{1}{6}x^3 + \frac{1}{120}x^5\right) dx$

$= \left[\frac{1}{2}x^2 - \frac{1}{24}x^4 + \frac{1}{720}x^6\right]_0^{1.5}$

$= \left(\frac{1}{2} \times 1.5^2 - \frac{1}{24} \times 1.5^4 + \frac{1}{720} \times 1.5^6\right) - (0)$

$= 0.93$ (2 DP)

31) d)

$\cos x \approx 1 - \frac{x^2}{2!} + \frac{x^4}{4!} = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

$\int_0^{1.5} \cos x \, dx \approx \int_0^{1.5} \left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4\right) dx$

$= \left[x - \frac{1}{6}x^3 + \frac{1}{120}x^5\right]_0^{1.5}$

$= \left(1.5 - \frac{1}{6} \times 1.5^3 + \frac{1}{120} \times 1.5^5\right) - (0)$

$= 1.00$ (2 DP)

31) e)

$(1-x)^{\frac{1}{2}} \approx 1 - \frac{x}{2} - \frac{x^2}{8}$

$\int_0^{0.5} (1-x)^{\frac{1}{2}} dx \approx \int_0^{0.5} \left(1 - \frac{x}{2} - \frac{x^2}{8}\right) dx$

$= \left[x - \frac{1}{4}x^2 - \frac{1}{24}x^3\right]_0^{0.5}$

$= \left(0.5 - \frac{1}{4} \times 0.5^2 - \frac{1}{24} \times 0.5^4\right) - (0)$

$= 0.43$ (2 DP)

32)

$(1-x)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right)(-x) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(-x)^2}{2} + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)(-x)^3}{6}$

$= 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3$

$\therefore \int \frac{1}{\sqrt{1-x}} dx \approx \int \left(1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3\right) dx = x + \frac{1}{4}x^2 + \frac{1}{8}x^3 + \frac{5}{64}x^4 + c$

33) a)

$\ln(1+2x) \approx (2x) - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} = 2x - 2x^2 + \frac{8}{3}x^3$

33) b)

$2x - 2x^2 + \frac{8}{3}x^3 = 8x^2$ gives $\frac{8}{3}x^3 - 10x^2 + 2x = 0$

$x(4x^2 - 15x + 3) = 0$ so $x = 0$ or $x = \frac{15}{8} \pm \frac{\sqrt{177}}{8}$

Since only valid for $-\frac{1}{2} < x \leq \frac{1}{2}$

So $x = 0.212$ (3 SF)

33) c)

Include the quartic term in the expansion, as every term included improves the accuracy.

34) a)

$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots$

$\approx 1 - \frac{1}{2}\theta^2 + \frac{1}{24}\theta^4$

34) b)

$-\frac{1}{648}\theta^4 = 1 - \frac{1}{2}\theta^2 + \frac{1}{24}\theta^4$ gives $7\theta^4 - 81\theta^2 + 162 = 0$

$(\theta^2 - 9)(7\theta^2 - 18) = 0$

So solutions are $\theta = 3$, $\theta = -3$, $\theta = \pm\sqrt{\frac{18}{7}}$

So $\theta = 1.60$ (3 SF)

35) a)

$f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots - \left(1 - \frac{x^2}{2!} - \cdots\right)$

$= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots - 1 + \frac{1}{2}x^2 + \cdots$

$\approx x + x^2 + \frac{1}{6}x^3$

35) b)

$x + x^2 + \frac{1}{6}x^3 = 0$

$\frac{1}{6}x(x^2 + 6x + 6) = 0$

Has solutions $x = 0, -3 \pm \sqrt{3}$

Therefore $x = -1.27$ (3 SF)

36) a)

$\ln\left(\frac{1+x}{1-x}\right) = \ln(1 + x) - \ln(1 - x)$

$\ln(1 + x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots$

$\ln(1 - x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \cdots$

$\ln\left(\frac{x+1}{1-x}\right) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots - \left(-x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \cdots\right)$

$= 2x + \frac{2}{3}x^3 + \cdots$

When $x = 0.2$

$\ln\left(\frac{1+0.2}{1-0.2}\right) = \ln\left(\frac{1.2}{0.8}\right) = \ln(1.5) = 2(0.2) + \frac{2}{3}(0.2)^3 + \cdots$

$= 0.405333 \ldots$

So 0.405 (3 d.p.)

36) b)

$\frac{\ln 1.5 - 0.405}{\ln 1.5} \times 100 = 0.115\%$

37) a)

$e^x\sqrt{1 + 2x} = e^x(1 + 2x)^{\frac{1}{2}} = \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)\left(1 + \left(\frac{1}{2}\right)(2x) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)(2x)^2}{2} + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(2x)^3}{6} + \cdots\right)$

$= \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots\right)\left(1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3 + \cdots\right)$

$= 1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3 + \cdots + x + x^2 - \frac{1}{2}x^3 + \cdots + \frac{1}{2}x^2 + \frac{1}{2}x^3 - \cdots + \frac{1}{6}x^3 \ldots$

$\approx 1 + 2x + x^2 + \frac{2}{3}x^3$

37) b)

$1 + 2(0.5) + (0.5)^2 + \frac{2}{3}(0.5)^3 = 2.33$ (3 SF)

37) c)

$\frac{\sqrt{2}\,e^{\frac{1}{2}} - 2.33}{\sqrt{2}\,e^{\frac{1}{2}}} \times 100 = 0.0705\%$

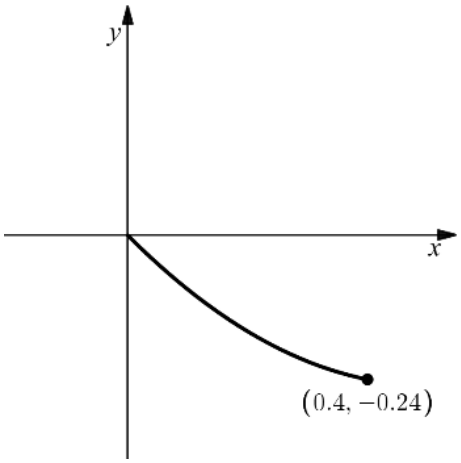
38)

$e^{-x} \approx 1 - x + \frac{x^2}{2!}$

$\cos x \approx 1 - \frac{x^2}{2!}$

$y = 1 - x + \frac{1}{2}x^2 - \left(1 - \frac{1}{2}x^2\right) = -x + x^2 = x(x-1)$

When $x = 0.4$, $y = 0.4 \times -0.6 = -0.24$



39)

The sum of two functions uniquely one of which must either be e^x or $\cos x$ as it goes through $(0, 1)$

So consider a point after $x \approx 2.3376$, e.g. $x = 2.4$ for the possible functions

But this is not valid for $\ln(1 + x)$ as $-1 < x \leq 1$

	4th term	$x = 2.4$	Sum of first 4 terms
e^x	$+\frac{x^3}{3!}$	2.304	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} = 8.584$
$\cos x$	$-\frac{x^6}{6!}$	-0.2654	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} = -0.7630208\ldots$
$\sin x$	$-\frac{x^7}{7!}$	-0.091	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} = 0.668550289\ldots$

$\therefore \cos x + \sin x$ is the only negative sum

So $\cos x$ and $\sin x$ must be $f(x)$ and $g(x)$

40)

Area is $\frac{1}{2}ab\sin C = \frac{1}{2} \times 2 \times 3 \times \sin(2x)$

$\sin 2x = 2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \cdots \approx 2x - \frac{4}{3}x^3$

Area is $\frac{1}{2} \times 2 \times 3 \times \left(2x - \frac{4}{3}x^3\right) = 6x - 4x^3$

$p = 6$ and $q = 4$

41) a)

$\cos x = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + \cdots$

$(1 + x)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right)x + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)x^2}{2} + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)x^3}{6} + \cdots$

$= 1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{5}{16}x^3 + \cdots$

$\frac{\cos x}{\sqrt{1+x}} = \left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + \cdots\right)\left(1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{5}{16}x^3 + \cdots\right)$

$= 1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{5}{16}x^3 + \cdots - \frac{1}{2}x^2 + \frac{1}{4}x^3 - \cdots$

$\approx 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{16}x^3$

41) b)

$y \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{16}x^3$

$\frac{dy}{dx} \approx -\frac{1}{2} - \frac{1}{4}x - \frac{3}{16}x^2$

When $x = 0$, $y = 1$

$\left.\frac{dy}{dx}\right|_{x=0} = -\frac{1}{2}$

$y = -\frac{1}{2}x + 1$

42) a)

$\sin(2x) = 2x - \frac{(2x)^3}{3!} + \cdots = 2x - \frac{4}{3}x^3 + \cdots$

$\sin^2(2x) = \left(2x - \frac{4}{3}x^3 + \cdots\right)\left(2x - \frac{4}{3}x^3 + \cdots\right)$

$= 4x^2 - \frac{8}{3}x^4 + \cdots - \frac{8}{3}x^3 + \cdots$

$\approx 4x^2 - \frac{16}{3}x^4$

42) b)

$16x^3 = 4x^2 - \frac{16}{3}x^4$ gives $4x^4 + 12x^3 - 3x^2 = 0$

$x^2(4x^2 + 12x - 3) = 0$

So $x = 0$, $x = -\frac{3}{2} \pm \sqrt{3}$

$\sin^2(2x) \geq 0$

$x = 0.232$ (3 SF)